

SIGNALS & SYSTEMS

2025 End-Semester Examination — Detailed, Easy-to-Read Solutions

Course	Signals and Systems
Course Code	216U41C404
Exam	End-Semester Examination
Date on paper	08 May 2025
Maximum Marks	50
Duration	2 Hours

How to use this PDF

- Read the boxed final answer first, then study the working.
- For numerical questions, write the governing formula before substitution.
- For transform questions, write the ROC/causality condition whenever relevant.
- The solutions follow the wording and notation of the uploaded 2025 paper.

Source: K. J. Somaiya School of Engineering, Signals and Systems ESE paper dated 08 May 2025.

Q1 — Short Questions + Difference Equation

Q1(i) says "Attempt any two"; all subparts are solved here for revision.

Q1(i)(a) Simplify $[e^{-t} \cos(3t - 60^\circ)]\delta(t)$

Use the sifting property $g(t)\delta(t)=g(0)\delta(t)$.

$$g(0)=e^0 \cos(-60^\circ)=1/2$$

$$\text{Therefore, } [e^{-t} \cos(3t-60^\circ)]\delta(t)=\frac{1}{2}\delta(t).$$

Q1(i)(b) Evaluate $\int_{-\infty}^{+\infty} \delta(\tau)f(t-\tau)d\tau$

The delta function samples its multiplier at $\tau=0$.

$$\int_{-\infty}^{+\infty} \delta(\tau)f(t-\tau)d\tau=f(t)$$

Final: $f(t)$.

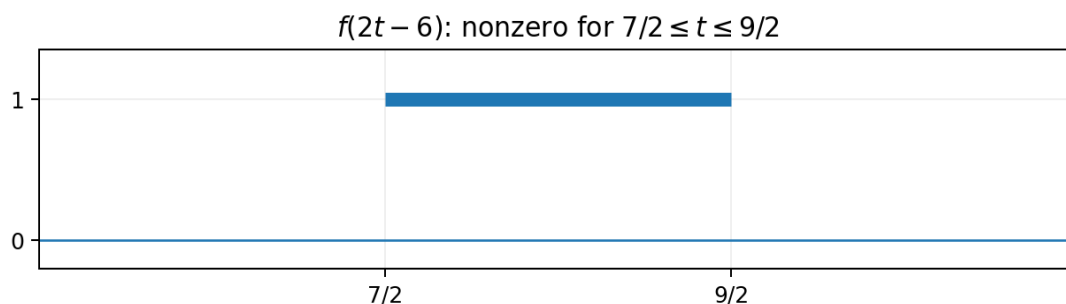
Q1(i)(c) Show that $\int_{-\infty}^{+\infty} \delta(t)e^{-j\omega t}dt=1$

$$\int_{-\infty}^{+\infty} \delta(t)e^{-j\omega t}dt=e^0=1$$

$$\text{Hence, } \int \delta(t)e^{-j\omega t}dt=1.$$

Q1(i)(d) Given $f(t)=1$ for $1 \leq t \leq 3$ and 0 otherwise. Find $f(2t-6)$.

$$1 \leq 2t-6 \leq 3 \rightarrow 7/2 \leq t \leq 9/2$$



Final: $f(2t-6)=1$ for $7/2 \leq t \leq 9/2$; 0 otherwise.

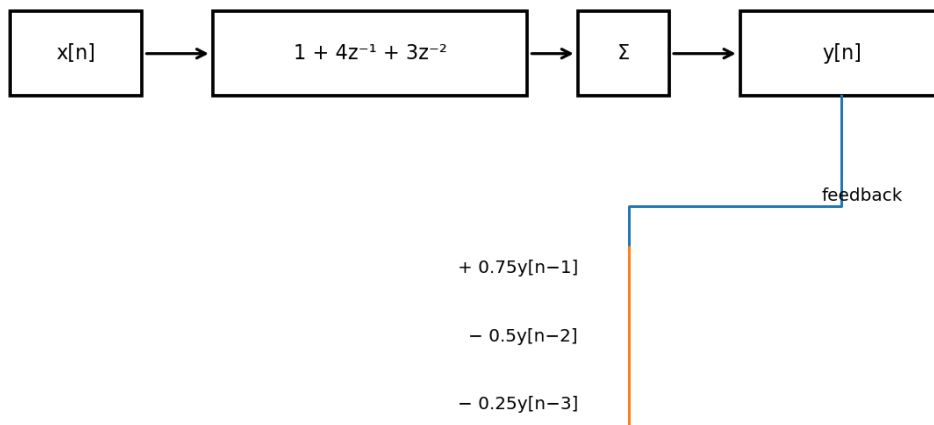
Q1(i)(e) Give the mathematical expression of any one even signal.

An even signal satisfies $x(t)=x(-t)$.

$$\text{Example: } x(t)=\cos(\omega_0 t).$$

Q1(ii) Find the output in the form of a difference equation.

The input-side path gives $x[n]+4x[n-1]+3x[n-2]$. The feedback gains are $+0.75$, -0.5 and -0.25 on successive delayed outputs.



$$y[n] = x[n] + 4x[n-1] + 3x[n-2] + 0.75y[n-1] - 0.5y[n-2] - 0.25y[n-3]$$

$$\text{Standard form: } y[n] - 0.75y[n-1] + 0.5y[n-2] + 0.25y[n-3] = x[n] + 4x[n-1] + 3x[n-2].$$

Q1(iii) Use Z-transform to perform the convolution.

$$h[n] = (1/2)^n, 0 \leq n \leq 2 \rightarrow \{1, 1/2, 1/4\}; x[n] = \delta[n] + \delta[n-1] + 4\delta[n-2]$$

$$H(z) = 1 + (1/2)z^{-1} + (1/4)z^{-2}; X(z) = 1 + z^{-1} + 4z^{-2}$$

$$Y(z) = X(z)H(z) = 1 + (3/2)z^{-1} + (19/4)z^{-2} + (9/4)z^{-3} + z^{-4}$$

$$\text{Final: } y[n] = \delta[n] + (3/2)\delta[n-1] + (19/4)\delta[n-2] + (9/4)\delta[n-3] + \delta[n-4].$$

Q2 — LTI Discrete Systems and Z-Transform

Q2(i) Given $x[n]$ and $y[n]$, find $H(z)$.

$$x[n] = (1/2)^n u[n] + 2^n u[n-1]; \quad y[n] = 6(1/2)^n u[n] - 6(3/4)^n u[n]$$

Use $Z\{a^n u[n]\} = z/(z-a)$. For $2^n u[n-1]$, the $n=0$ sample is absent.

$$X(z) = z/(z-1/2) + 2/(z-2) = (z^2-1)/[(z-1/2)(z-2)]$$

$$Y(z) = 6z/(z-1/2) - 6z/(z-3/4) = -(3/2)z/[(z-1/2)(z-3/4)]$$

$$H(z) = Y(z)/X(z) = -(3/2)z(z-2)/[(z-3/4)(z^2-1)]$$

$$\text{Final: } H(z) = -(3/2)z(z-2)/[(z-3/4)(z^2-1)].$$

Q2(ii) Find $H(z)$ and $h[n]$ for the causal LTI system.

$$y[n] = 0.25y[n-1] + 0.125y[n-2] + x[n] - x[n-1]$$

$$Y(z) = 0.25z^{-1}Y(z) + 0.125z^{-2}Y(z) + X(z) - z^{-1}X(z)$$

$$H(z) = Y(z)/X(z) = (1-z^{-1})/(1-0.25z^{-1}-0.125z^{-2})$$

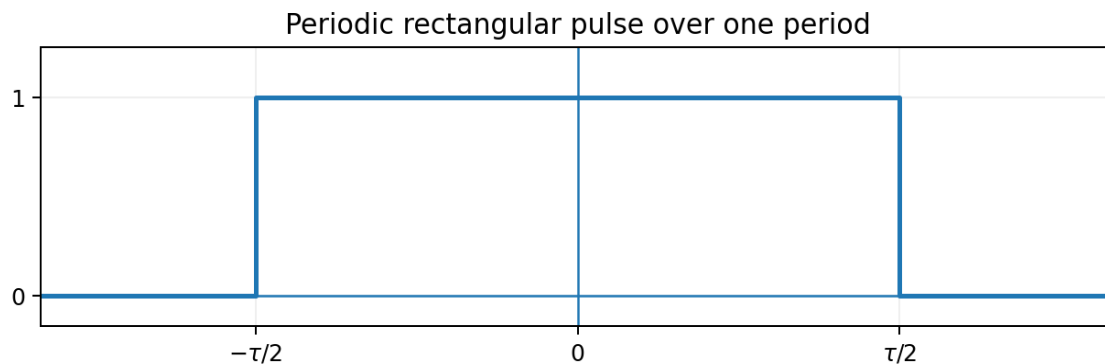
$$= (1-z^{-1})/[(1-0.5z^{-1})(1+0.25z^{-1})]$$

$$H(z) = (5/3)/(1+0.25z^{-1}) - (2/3)/(1-0.5z^{-1})$$

$$\text{For the causal system: } h[n] = (5/3)(-1/4)^n u[n] - (2/3)(1/2)^n u[n].$$

Q3 — Fourier Series and Fourier Transform

Q3(i) Find the Fourier series of the periodic rectangular pulse.



$f(t)=1$ for $|t|\leq\tau/2$; 0 otherwise, repeated every T ; $\omega_0=2\pi/T$

$f(t)=a_0/2+\sum_{n=1}^{\infty}[a_n\cos(n\omega_0t)+b_n\sin(n\omega_0t)]$

The signal is even, so $b_n=0$.

$a_0=(2/T)\int_{-\tau/2}^{\tau/2}dt=2\tau/T$

$a_n=(2/T)\int_{-\tau/2}^{\tau/2}\cos(n\omega_0t)dt=(2/(n\pi))\sin(n\pi\tau/T)$

Therefore, $f(t)=\tau/T+\sum_{n=1}^{\infty}[2/(n\pi)\cdot\sin(n\pi\tau/T)]\cos(2\pi nt/T)$.

Exponential form:

$C_0=\tau/T$, $C_k=\sin(k\pi\tau/T)/(k\pi)$, $k\neq 0$

Equivalently, $C_k=(\tau/T)\text{sinc}(k\pi\tau/T)$, where $\text{sinc}(x)=\sin(x)/x$.

Q3(ii) Obtain the Fourier transform of the rectangular pulse.

$g_{\tau}(t)=1$ for $|t|\leq T/2$; 0 otherwise

$G(j\omega)=\int_{-T/2}^{T/2}e^{-j\omega t}dt=2\sin(\omega T/2)/\omega=T \text{ sinc}(\omega T/2)$

Final: $G(j\omega)=2\sin(\omega T/2)/\omega=T\cdot\text{sinc}(\omega T/2)$, and $G(0)=T$.

Q4 — Inverse Laplace Transform

$$F(s) = (s+3+5e^{-2s})/[(s+1)(s+2)]$$

$$F(s) = (s+3)/[(s+1)(s+2)] + 5e^{-2s}/[(s+1)(s+2)]$$

Partial fractions for the first term:

$$(s+3)/[(s+1)(s+2)] = 2/(s+1) - 1/(s+2)$$

$$L^{-1}\{(s+3)/[(s+1)(s+2)]\} = 2e^{-t} - e^{-2t}$$

For the delayed term use $L^{-1}\{e^{-as}F(s)\} = u(t-a)f(t-a)$.

$$L^{-1}\{1/[(s+1)(s+2)]\} = e^{-t} - e^{-2t}$$

$$5e^{-2s}/[(s+1)(s+2)] \rightarrow 5u(t-2)[e^{-(t-2)} - e^{-2(t-2)}]$$

$$\text{Final: } f(t) = 2e^{-t} - e^{-2t} + 5u(t-2)[e^{-(t-2)} - e^{-2(t-2)}].$$

Q5 — Discrete-Time Convolution

$$x[n] = (1/6)^n u[n], \quad h[n] = (1/3)^n u[n-3]$$

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k]$$

$$y[n] = \sum_k (1/6)^k u[k] (1/3)^{n-k-3} u[n-k-3]$$

The unit steps require $0 \leq k \leq n-3$. Thus $y[n] = 0$ for $n < 3$.

$$\text{For } n \geq 3: y[n] = \sum_{k=0}^{n-3} (1/6)^k (1/3)^{n-k-3}$$

$$= (1/3)^{n-3} \sum_{k=0}^{n-3} (1/2)^k$$

$$= 2(1/3)^{n-3} [1 - (1/2)^{n-2}]$$

$$\text{Final: } y[n] = 2(1/3)^{n-3} [1 - (1/2)^{n-2}] u[n-3].$$

$$\text{Equivalent: } y[n] = [54(1/3)^n - 216(1/6)^n] u[n-3]$$

Check: $y[3] = x[0]h[3] = 1$, matching the formula.

One-Page Last-Minute Formula Sheet

Topic	Formula
Delta sifting	$\int x(t)\delta(t-t_0)dt = x(t_0)$
Z-transform	$Z\{x[n]\} = \sum x[n]z^{-n}$
Right-sided exponential	$Z\{a^n u[n]\} = z/(z-a)$, ROC $ z > a $
Convolution	$y[n] = \sum x[k]h[n-k]$
LTI system	$H(z) = Y(z)/X(z)$
Difference equation	Take Z-transform assuming initial rest
CT Fourier series	$f(t) = a_0/2 + \sum [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$
CTFT	$X(j\omega) = \int x(t)e^{-j\omega t}dt$
Rectangular pulse FT	$2\sin(\omega T/2)/\omega = T \text{ sinc}(\omega T/2)$
Laplace time shift	$L^{-1}\{e^{-as}F(s)\} = u(t-a)f(t-a)$

Exam-writing checklist

- State the formula/property first.
- For Z-transform problems, write the ROC/causality condition.
- For Fourier series, use symmetry before calculating coefficients.
- For convolution, clearly state the valid index limits.
- Box the final result and keep index shifts consistent.